SQL Basics

Q1. Create a New Database and Table for Employees

Step 1: Create a New Database

```
CREATE DATABASE company_db;
```

Step 2: Select the Database

```
USE company_db;
```

Step 3: Create the Employees Table

```
CREATE TABLE employees (
    employee_id  INT          PRIMARY KEY,
    first_name   VARCHAR(50),
    last_name    VARCHAR(50),
    department   VARCHAR(50),
    salary       INT,
    hire_date    DATE
);
```

Step 4: Verify the Table

Check that the table was created successfully:

```
SHOW TABLES;        -- lists all tables in the database
DESCRIBE employees; -- shows column names and data types
```

Q2. Insert Data into the Employees Table

Insert Multiple Records

```
INSERT INTO employees (employee_id, first_name, last_name, department,
salary, hire_date)
VALUES
    (101, 'Amit',  'Sharma', 'HR',      50000, '2020-01-15'),
    (102, 'Riya',  'Kapoor', 'Sales',   75000, '2019-03-22'),
    (103, 'Raj',   'Mehta',  'IT',      90000, '2018-07-11'),
    (104, 'Neha',  'Verma',  'IT',      85000, '2021-09-01'),
    (105, 'Arjun', 'Singh',  'Finance', 60000, '2022-02-10');
```

Verify Inserted Data

```
SELECT * FROM employees;
```

Q3. Display All Employee Records Sorted by Salary (Low → High)

Use ORDER BY salary to sort results in ascending order (default):

```
SELECT *
FROM   employees
ORDER BY salary ASC;
```

Q4. Show Employees Sorted by Department (A–Z) then Salary (High → Low)

Multiple sort columns can be chained in ORDER BY:

```
SELECT *
FROM   employees
ORDER BY department ASC,
         salary     DESC;
```

Q5. List All IT Employees Ordered by Hire Date (Newest First)

Combine WHERE to filter and ORDER BY to sort:

```
SELECT *
FROM   employees
WHERE  department = 'IT'
ORDER BY hire_date DESC;
```

Q6. Create and Populate a Sales Table

Step 1: Create the Sales Table

```
CREATE TABLE sales (
    sale_id       INT          PRIMARY KEY,
    customer_name VARCHAR(50),
    amount        INT,
    sale_date     DATE
);
```

Step 2: Insert Records

```
INSERT INTO sales (sale_id, customer_name, amount, sale_date)
VALUES
    (1, 'Aditi', 1500, '2024-08-01'),
    (2, 'Rohan', 2200, '2024-08-03'),
    (3, 'Aditi', 3500, '2024-09-05'),
    (4, 'Meena', 2700, '2024-09-15'),
    (5, 'Rohan', 4500, '2024-09-25');
```

Step 3: Verify Data

```
SELECT * FROM sales;
```

Q7. Display All Sales Records Sorted by Amount (High → Low)

```
SELECT *
FROM   sales
ORDER BY amount DESC;
```

Q8. Show All Sales Made by Customer "Aditi"

```
SELECT *
FROM   sales
WHERE  customer_name = 'Aditi';
```

Q9. Difference Between a Primary Key and a Foreign Key

Feature	Primary Key	Foreign Key
Definition	Uniquely identifies each record in a table	Refers to a Primary Key in another table
Uniqueness	Must be unique	Can have duplicate values
NULL Values	NOT allowed	Can be NULL (in most cases)
Purpose	Ensures each row is uniquely identifiable	Maintains referential integrity between tables
Per Table	Only one Primary Key (can be composite)	A table can have multiple Foreign Keys
Example	employee_id in the employees table	employee_id in a salary table referencing employees

Q10. What Are Constraints in SQL and Why Are They Used?

Definition

Constraints are rules applied to table columns that control the type of data allowed. They enforce data accuracy, consistency, and integrity throughout the database.

Why Are Constraints Used?

- To prevent invalid or incomplete data from being inserted
- To maintain data integrity and reliability
- To enforce business rules at the database level
- To define and preserve relationships between tables

Types of Common Constraints

Constraint	Description
PRIMARY KEY	Uniquely identifies each record; does not allow NULL or duplicate values
FOREIGN KEY	Links a column in one table to the Primary Key of another table
NOT NULL	Ensures a column cannot store a NULL value
UNIQUE	Ensures all values in a column are distinct (allows one NULL)
CHECK	Restricts values in a column based on a specified condition
DEFAULT	Assigns a default value to a column when no value is provided on insert

All my practical work done here:
https://drive.google.com/drive/u/0/folders/1Zaj0t1_FaLZwSUFVIE5N31Bg_AfkOALI